

blinkit DARK STORE OPERATIONS ANALYSIS

Loss Attribution & Efficiency Insights



UNDERSTAND
Key Loss Drivers



OPTIMIZE
Store Operations



IMPACT
Reduce Losses



Business Problem & Key Findings

The Operational Challenge

Dark store operations generate financial losses through inventory damage, handling failures, expiry, packaging issues, and storage-related incidents. These losses are often preventable — but without systematic attribution, they remain invisible to decision-makers.

Over the 90-day analysis window (April 24 – July 23, 2026), Blinkit's dark stores recorded **₹532K in total losses** across **200 incidents**, damaging **2,067 units**. The critical finding: **73% of losses are classified as preventable**.

Key Findings at a Glance

₹532K total recorded loss across 200 incidents

73% of losses classified as preventable

41.3% of loss concentrated in top 3 damage drivers

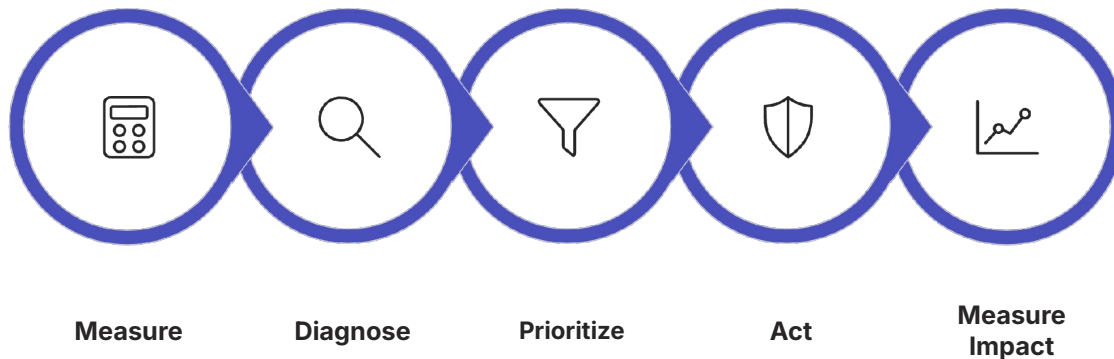
Store S001 contributes 26.22% of total network loss

Handling Error is the largest individual loss driver

₹42K monthly / ₹501K annual savings opportunity under 35% target-reduction scenario

Business Problem & Strategic Objectives

Without systematic loss attribution, dark store operations absorb recurring financial damage without clear accountability or targeted intervention. This analysis provides the operational intelligence needed to shift from reactive loss absorption to proactive loss prevention.



This structured workflow ensures every analytical finding translates directly into an actionable operational intervention — closing the gap between insight and impact.

01

Quantify Financial Losses

Measure total loss exposure across all incident types and stores.

02

Identify Highest-Value Damage Drivers

Rank loss categories by financial contribution and frequency.

03

Determine Preventable Loss Share

Classify which losses can realistically be eliminated through operational change.

04

Identify High-Exposure Stores & Roles

Pinpoint where losses concentrate to focus intervention resources.

05

Estimate Financial Impact of Target Reduction

Translate the 35% reduction target into a concrete savings opportunity.

Data Sources & Analytical Approach

Dataset Inventory

The analysis draws on five integrated datasets from Blinkit's operational data layer, covering the full incident lifecycle from product master to employee performance.

Dataset	Records
Damaged Products	200
Employee Performance	40
Employee Master	40
Products Master	150
Putaway Transactions	500

Tools & Technologies

Python (Pandas, NumPy), Matplotlib, Seaborn, and Power BI — ensuring reproducibility and executive-ready visualization.

Analytical Methods Applied

- Data quality assessment and validation against Power BI source of truth
- Loss attribution by damage reason, store, and employee role
- Severity classification and preventable-loss tagging
- Pareto analysis to identify the vital few loss drivers
- Frequency vs. cost opportunity matrix for prioritization
- Savings scenario modeling under the 35% target-reduction framework

❏ The Power BI dashboard is the authoritative source for all executive KPIs. Notebook figures serve as supporting analytical context only.

Total Financial Exposure: ₹532K Across 200 Incidents

₹532K

Total Recorded Loss

Across all dark stores during the analysis window

200

Total Incidents

Operational incidents recorded

2,067

Units Damaged

Individual product units affected

₹2,659

Avg. Loss / Incident

Mean financial impact per incident

₹163K

Avg. Monthly Loss

Monthly run-rate of operational losses

₹1.96M

Annual Loss Projection

Extrapolated full-year exposure at current rate

Loss Concentration

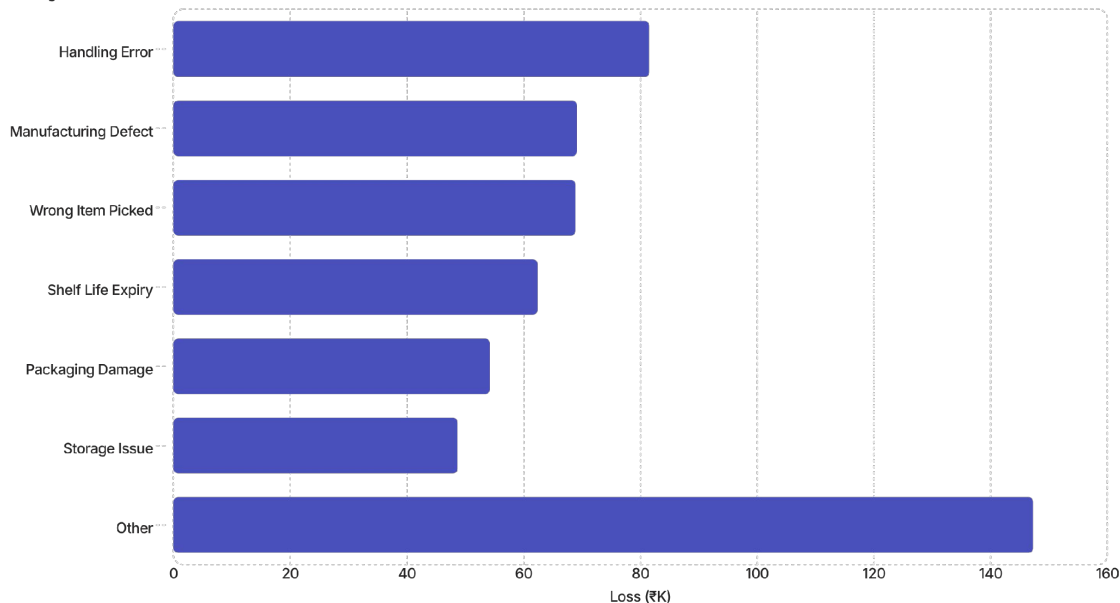
Store S001 alone accounts for **26.22% of total network loss**, making it the highest-priority intervention target. The top three stores collectively represent a disproportionate share of financial exposure, supporting a focused, store-tiered prevention strategy rather than a uniform network-wide rollout.

Monthly Loss Trajectory

With an average monthly loss of **₹163K**, the data reveals a consistent loss pattern without significant seasonal dampening — indicating structural, not episodic, operational issues that require systematic intervention.

Top 3 Damage Drivers Account for 41.3% of Total Loss

Damage Driver



Handling Error — ₹81.51K

The largest individual loss driver. Process controls, SOP adherence, and targeted training represent the primary prevention levers. This category warrants immediate operational review.

Manufacturing Defect — ₹69.12K

While partially external to dark store operations, supplier quality agreements and inbound inspection protocols can reduce downstream financial impact.

Wrong Item Picked — ₹68.88K

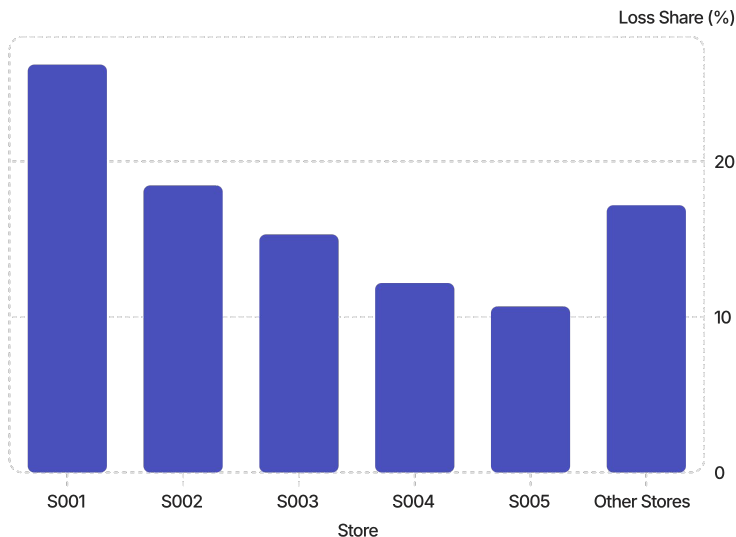
A process and training issue with clear intervention paths: pick-verification workflows, barcode scanning, and role-specific training modules.



These three drivers combined represent 41.3% of total loss — a concentrated opportunity for targeted intervention. Shelf Life Expiry, while not in the top 3 by total value, represents a high cost-per-incident prevention opportunity.

Loss Concentration: Store S001 & Severity Distribution

Store-Level Loss Contribution



Store S001 contributes **26.22%** of total network loss — nearly double the share of the next-highest store. A tiered intervention approach should prioritize S001 for immediate operational review and root-cause investigation.

Severity Classification

Of the 200 incidents recorded, **1.41% are classified as high-severity** — representing outsized financial impact per event. While low in frequency, high-severity incidents warrant dedicated investigation protocols given their disproportionate cost contribution.

High-Severity

1.41% of incidents — highest cost per event

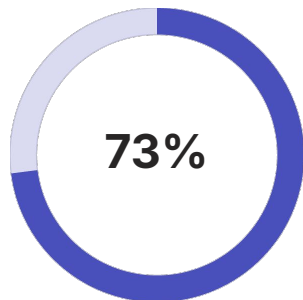
Medium-Severity

Moderate frequency with meaningful financial impact

Low-Severity

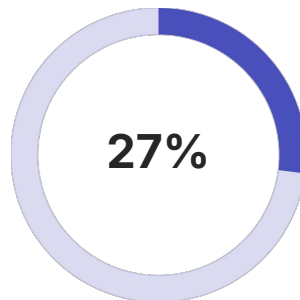
High frequency, lower individual cost — volume-driven

73% of Losses Are Preventable — The Operational Workforce Connection



Preventable Loss Share

Of total financial loss is classified as preventable through operational intervention



Non-Preventable

Losses from external factors (e.g., manufacturing defects) with limited dark-store control

What "Preventable" Means in Practice

Preventable losses are those attributable to operational processes within Blinkit's direct control — including handling, picking accuracy, storage compliance, packaging standards, and shelf-life management. This 73% figure represents the addressable opportunity surface for targeted intervention programs.

Employee & Role-Level Insights

Analysis of the 40-employee performance dataset reveals variation in incident rates across operational roles. Roles with the highest incident-to-transaction ratios represent priority targets for training and process reinforcement. Reporting-delay analysis further identifies opportunities to accelerate incident documentation and root-cause capture.

- Employee-level findings should be used for coaching and process improvement — not punitive action. The goal is systemic improvement, not individual attribution.

₹501K Annual Savings Opportunity Under the 35% Target-Reduction Scenario

₹42K

Potential Monthly Savings

Estimated monthly reduction under the target scenario

35%

Target Reduction

Management-defined loss reduction goal

₹501K

Annual Savings Opportunity

Full-year financial benefit at 35% loss reduction

₹257.2

Loss Per Damaged Unit

Average financial cost per unit damaged

Scenario Modeling Framework

The savings estimates are derived from a **35% target-reduction scenario** applied to the preventable loss base (73% of ₹532K). This is a planning model — not a guaranteed outcome. Actual savings will depend on intervention quality, adoption rates, and operational execution.

- Preventable loss base: ~₹388K (73% of ₹532K)
- 35% reduction target applied to preventable base
- Monthly run-rate: ₹42K estimated savings
- Annual projection: ₹501K estimated opportunity

Key Financial Metrics

Metric	Value
Total Loss	₹532K
Average Units / Incident	10.3
Loss / Damaged Unit	₹257.2
Average Monthly Loss	₹163K
Annual Loss Projection	₹1.96M
Potential Monthly Savings	₹42K
Annual Savings Opportunity	₹501K

Four Priority Intervention Areas

The following recommendations are prioritized by financial impact, preventability, and implementation feasibility. Each area maps directly to the loss drivers identified in this analysis.

1

Handling Error Reduction Program

As the largest individual loss driver (₹81.51K), Handling Error warrants immediate investment in SOP reinforcement, hands-on training modules, and handling-quality audits — particularly at Store S001.

2

Picking Accuracy & Verification Workflow

Wrong Item Picked incidents (₹68.88K) can be reduced through barcode scanning at pick and pack stages, dual-verification for high-value SKUs, and role-specific picking accuracy training.

3

Shelf Life & Expiry Management

Shelf Life Expiry represents a high cost-per-incident opportunity. Implement FEFO (First Expiry, First Out) protocols, automated expiry alerts, and daily shelf-audit checklists to reduce write-off frequency.

4

Store S001 Deep-Dive Investigation

With 26.22% of total network loss, Store S001 requires a dedicated root-cause investigation — examining staffing patterns, process adherence, layout constraints, and incident reporting quality to identify store-specific intervention levers.

- ☐ **Next Steps:** Establish a monthly loss-attribution review cadence, assign ownership of each intervention area, and track progress against the 35% reduction target using the Power BI dashboard as the single source of truth.